

Synthesis and characterization of a chiral Zn(II) complex based on a *trans*-1,2-diaminocyclohexane derivative and catalytic reduction of acetophenone

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Abstract

Reaction of ZnCl_2 with *R,R*-bmbed (= *N,N'*-bis(4-methoxy-benzyl)-cyclohexane-1,2-diamine) **1** in ethanol yields the chiral $[\text{ZnCl}_2(\text{R,R-bmbed})]$ **2**, which crystallizes in the monoclinic space group $P2_1$. The Zn atom has a distorted tetrahedral geometry involving two nitrogen atoms of *R,R*-bmbed and two chloro ligands with the range of bond lengths, 2.066(2)–2.2289(9) Å. The result of the catalytic enantioselective reduction of acetophenone promoted by *R,R*-bmbed: $\text{Zn}(\text{OTf})_2 = 1:1$ mole ratio in the presence of catecholborane for 48 h at 0 °C give (*S*)-(–)-1-phenylethanol in 48% ee with 77% yield.

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1. Introduction

Catalytic asymmetric synthesis is a key technique for preparing optically active substances in biologically medical science and fine chemicals [1]. Enantioselective syntheses such as the alkylation of aldehydes [2], the epoxidation of olefines [3] and the asymmetric reduction of ketones [4] have been extensively studied. In particular, asymmetric alkylation of aldehydes with dialkylzinc in the presence of chiral bidentate ligands such as diol [5], aminoalcohol [6], diamine [7], ditriflamide [2,8], diaminocyclohexane [9] have been well-established as contrasted with the asymmetric reduction of ketones. The Kobayashi and co-workers [2a] has obtained highly optically pure alcohols by using a (*R,R*)-*trans*-1,2-diaminocyclohexane derivative containing an electron withdrawing group in the presence of $\text{Ti}(\text{O}^i\text{Pr})_4$. Recently, the Walsh and co-worker [10] has worked to optimize the enantioselectivity of the catalyst by varying the achiral ligands, described as chiral environment

amplifications, which is a modular approach to asymmetric catalysis. Enantioselective reduction of α -alkoxy ketones in the presence of $\text{Zn}(\text{OTf})_2$ and bis-oxazolines afforded the chiral alcohols in good yields and high ee's [11], but the extended studies for enantioselective reduction of α -alkoxy ketones are limited.

The enantiomeric purity of a chiral auxiliary in asymmetric catalysis is a major factor for alkylation of benzaldehyde. The reaction rate is also considerably influenced by the enantiomeric purity, and is significantly faster with pure chiral auxiliaries than with the racemic ones [12]. The structure of the catalyst in the asymmetric reduction of ketones can be a key parameter for the development of a better catalyst, providing the coordination environments at the centre atom, the reaction mechanism and the steric effect. Diaminocyclohexane derivatives involving functional groups such as hydrogen, isopropyl, isobutyl, etc., are not satisfied with the enantioselective reduction of ketones by comparing with the enantioselective reduction of aldehydes. However, the enantioselective reduction of acetophenone to secondary alcohols with polymethylhydrosiloxane (PMHS) was carried out with high yields and ee's in the

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presence of chiral zinc catalysts [13] or chiral titanocene [14]. More recently, we have reported our results for the catalytic reduction of acetophenone using the $\text{Zn}(\text{OTf})_2$ and N,N' -bis(3,5-dimethylpyrazolylmethyl)-L-alanine-methylester system, in which (*S*)-(-)-1-phenylethanol was obtained in low yield and ee [15].

Optically pure 1,2-diaminocyclohexane derivatives have been extensively utilized as chiral reagents, auxiliary catalysts and chiral ligands in various asymmetric synthetic routes [16]. However, these ligands are rare in the asymmetric reduction of ketones. The studies of the enantiomeric reduction of acetophenone activated by diaminocyclohexane derivatives bearing secondary and tertiary diamine bidentate ligands under mild conditions have not been reported. Herein, we have synthesized optically pure *R,R*-bmbcd and its complex of ZnCl_2 , and describe its catalytic reduction of acetophenone by considering its structural property and the steric effect through the function of the methoxy-benzyl group.

2. Experimental

2.1. General remark

All reagents were of reagent grade and solvents were freshly distilled prior to use. ^1H NMR spectra were recorded on a Varian 300-NMR Spectrometer at ambient temperature and chemical shifts were referenced to internal tetramethylsilane. HPLC with a chiralcel OD-H (0.46 cm $\Phi \times 25$ cm) column was used for measurement of ee. Elemental analyses were performed at the Chemical Analysis Laboratory of the Center for Scientific Instruments of Kyungpook National University.

2.2. Syntheses of the compounds

2.2.1. N,N' -bis(4-methoxy-benzyl)-cyclohexane-1,2-diamine (*R,R*-bmbcd), **1**

Absolute ethanol was added to *trans-R,R*-1,2-diaminocyclohexane (19.81 g, 0.17 mol) until it was completely dissolved. *p*-Anisaldehyde (43.077 ml, 0.347 mol) was added and the yellow solution was stirred at room temperature for two days. The solvent was then removed under reduced pressure to obtain a thin yellow solid. This solid was recrystallized in diethyl ether to give colorless crystals (55.00 g, 90%). To a solution of the colorless crystals an ice bath (16.81 g, 0.0478 mol) in dry THF, in an ice bath, was slowly added LiAlH_4 (1 M solution in THF, 143.904 ml, 3 equiv.) by syringe. This solution was refluxed for 24 h, cooled to room temperature and then distilled water (12 ml) and 2 M NaOH (30 ml) were added. After filtering, the solvent was removed under reduced pressure, the remaining solid was extracted with CH_2Cl_2 , dried with MgSO_4 , filtered, and then the solvent was evaporated to give a thin yellow solid. This solid was

recrystallized in diethyl ether to give colorless crystals (13.60 g, 80%). *Anal.* Calc. for $\text{C}_{22}\text{H}_{30}\text{N}_2\text{O}_2$: C, 74.54; H, 8.53; N, 7.90. Found: C, 74.57; H, 8.49; N, 7.87%. ^1H NMR (CDCl_3); δ 7.2 (d, 4H, Ph, $J = 8.7$ Hz), 6.8 (d, 4H, Ph, $J = 8.4$ Hz), 3.77 (s, 6H, OCH_3), 3.84–3.56 (dd, 4H, CH_2 , $J = 13.2$ Hz), 2.22 (m, 2H, CHA), 2.12 (m, 2H, CHA), 1.74 (m, 2H, CHA), 1.69 (m, 2H, CHA), 1.20 (m, 1H, CHA), 1.01 (m, 1H, CHA).

2.2.2. $\text{ZnCl}_2(\text{R,R-bmbcd})$, **2**

To a solution of ZnCl_2 (0.7900 g, 5.79 mmol) in ethanol (10 ml) was added dropwise a solution of **1** (2.0548 g, 5.796 mmol) in ethanol (25 ml). The mixture was stirred overnight at ambient temperature. The white precipitate that formed was filtered off and washed several times with *n*-hexane and diethyl ether. The white solid was recrystallized from methanol to obtain colorless crystals (2.625 g, 92%). *Anal.* Calc. for $\text{C}_{22}\text{H}_{30}\text{Cl}_2\text{N}_2\text{O}_2\text{Zn}$: C, 53.84; H, 6.16; N, 5.71. Found: C, 53.82; H, 6.16; N, 6.18%. ^1H NMR (CDCl_3); δ 7.4 (d, 4H, Ph, $J = 8.7$ Hz), 6.8 (d, 4H, Ph, $J = 8.4$ Hz), 3.73 (s, 6H, OCH_3), 3.99–3.82 (dd, 4H, CH_2 , $J = 13.2$ Hz), 2.46 (m, 4H, CHA), 2.08 (m, 2H, CHA), 1.85 (m, 2H, CHA), 1.26 (m, 1H, CHA), 1.13 (m, 1H, CHA).

2.3. Crystal structure determination of **2**

A colorless single crystal of **2** suitable for X-ray analysis was sealed in a thin walled glass capillary and mounted on an Enraf-Nonius CAD-4 diffractometer with Mo $\text{K}\alpha$ radiation ($\lambda = 0.71073$ Å). Unit cell dimensions with estimated standard deviations were determined by least-squares using 25 well-centered reflections in the range of 10° – 13° . A total of 2469 unique reflections were collected using the ω – 2θ scan mode. The intensities of the reflections were corrected for Lorentz and polarization effects. Crystal data, data collection, and refinement for the complex are listed in Table 1. Data reduction was carried out using a MOLEN program package; a decay correction based on the intensities of three standard reflections measured every hour was performed; empirical absorption corrections were applied based on Ψ scans [17]. The structure in $P2_1$ for the complex was determined by direct methods and refined by full-matrix least-squares using SHELXS-97 and SHELXL-97 program packages [18,19] using reflections with $I > 2\sigma(I)$. All non-hydrogen atoms were refined anisotropically. Hydrogen atoms were constrained by using riding models. The final cycle of the refinement yielded $R = 0.021$ and $wR = 0.057$. Selected bond distances and bond angles are given in Table 2.

2.4. Catalytic enantioselective reduction of acetophenone

The flame-dried Schlenk flask was charged with the chiral ligand **1** (0.078 mmol), $\text{Zn}(\text{OTf})_2$ (0.078 mmol) and

Table 1
Crystallographic data and structure refinement of **2**

Empirical formula	C ₂₂ H ₃₀ Cl ₂ N ₂ O ₂ Zn
Formula weight	490.75
Temperature (K)	293(2)
Wavelength, λ (Mo K α) (Å)	0.71073
Crystal system	monoclinic
Space group	<i>P</i> 2 ₁
<i>Unit cell dimensions</i>	
<i>a</i> (Å)	8.0309(5)
<i>b</i> (Å)	13.0098(7)
<i>c</i> (Å)	11.2650(6)
β (°)	99.092(5)
Volume <i>V</i> (Å ³)	1162.2(1)
Formula units per cell <i>Z</i>	2
<i>D</i> _{calc} (Mg/m ³)	1.402
Absorption coefficient (mm ^{−1})	1.306
<i>F</i> (000)	512
θ Range for data collection (°)	1.83–25.47
Index ranges	−9 ≤ <i>h</i> ≤ 9, 0 ≤ <i>k</i> ≤ 15, 0 ≤ <i>l</i> ≤ 13
Reflections collected/unique	2469/2259 [<i>R</i> _{int} = 0.0117]
Refinement method	full-matrix least-squares on <i>F</i> ²
Data/restraints/parameters	2259/1/263
Goodness-of fit on <i>F</i> ²	1.060
Final <i>R</i> indices [<i>I</i> > 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.021, <i>wR</i> ₂ = 0.057
Largest differential peak and hole (e/Å ³)	0.19 and 0.15

Table 2
Selected bond lengths (Å) and angles (°) for **2**

<i>Bond lengths</i>			
Zn–N(1)	2.066(2)	Zn–N(2)	2.092(2)
Zn–Cl(1)	2.2289(2)	Zn–Cl(2)	2.2036(8)
N(1)–C(8)	1.489(4)	N(1)–C(9)	1.486(3)
N(2)–C(14)	1.492(4)	N(2)–C(15)	1.499(4)
C(9)–C(14)	1.533(4)		
<i>Bond angles</i>			
N(1)–Zn–N(2)	85.66(9)	N(1)–Zn–Cl(1)	111.55(7)
N(1)–Zn–Cl(2)	113.32(7)	N(2)–Zn–Cl(1)	108.43(7)
N(2)–Zn–Cl(2)	110.60(7)	Cl(1)–Zn–Cl(2)	121.42(3)
N(1)–C(8)–C(5)	112.1(3)	N(2)–C(15)–C(16)	111.0(2)
C(8)–N(1)–C(9)	114.1(2)	C(14)–N(2)–C(15)	112.0(2)
C(8)–N(1)–Zn	115.3(2)	C(9)–N(1)–Zn	104.5(2)
C(14)–N(2)–Zn	106.2(2)	C(15)–N(2)–Zn	114.14(9)

dry CH₂Cl₂ (2 ml) under nitrogen atmosphere via syringe. The mixture was stirred for 30 min, cooled to 0 °C and then acetophenone (0.078 mmol) was added. The mixture was then stirred for 5 min and catecholborane (100 μ l, 0.94 mmol) was added. The procedures were same as the methods used in the previous paper [15]. The enantiomeric excess was measured by HPLC (a chiral OD column), affording (*S*)-(–)-1-phenylethanol in 48% ee with 77% yield.

3. Results and discussion

3.1. Synthesis and chemical properties

The pure optical material, diaminocyclohexane derivative **1**, which was afforded from the cyclohexyldii-

mine derivative by reduction with LiAlH₄, gives in quantitative yield a colorless solid. It is soluble in polar solvents and insoluble in non-polar solvents such as hydrocarbons.

The reaction of ZnCl₂ with **1** afforded **2** in a high yield as white crystals after the usual workups. The chiral Zn(II) complex is readily soluble in polar solvents such as alcohols, yet poorly soluble in most polar solvents such as chloroform and THF.

3.2. Crystal structure of complex **2**

The crystal structure of the chiral Zn(II) complex **2**, for getting binding information of the catalytic species, is confirmed by X-ray analysis as shown in Fig. 1. The central Zn atom is coordinated by two nitrogen atoms from a bidentate *R,R*-bmbcd and two chloro atoms in a distorted tetrahedral arrangement. The bond distances Zn–N(1) and Zn–N(2) are 2.066(2) and 2.092(2) Å, respectively, which are comparable to those in related compounds of the type ZnCl₂(L), i.e., 2.029(3) and 2.157(3) Å in ZnCl₂(Clemizole) where Clemizole is 1-(*p*-chloro-phenylmethyl)-2-(1-pyrrolidinylmethyl)benzimidazole [20], 2.006(3) and 2.008(3) Å in ZnCl₂(1,2-dimethylimidazole)₂ [21], and 2.050(3) and 2.074(3) Å in ZnCl₂(quinoline) [22]. The bond distances Zn–Cl(1) and Zn–Cl(2) [2.2289(9) and 2.2036 (8) Å] are close to those of similar complexes which are in the range of 2.196(2)–2.2509(8) Å [20–22]. The long bond distances of Zn–Cl are comparable with the bond distances of Zn–N, giving rise to those reported for the ZnCl₂N₂ core. The bond angles of N–Zn–Cl are in the range of 108.43(7)°–113.32(7)°, which are distorted from ideal tetrahedral geometry as a consequence of the opening of the two chloro ligands. The bond angle N(1)–Zn–N(4) [85.66(9)°] is smaller than that [89.5(1)°] found in ZnCl₂(2,2'-bis(pyrazolyl)-1-ylpropane) and that [109.3(1)°]

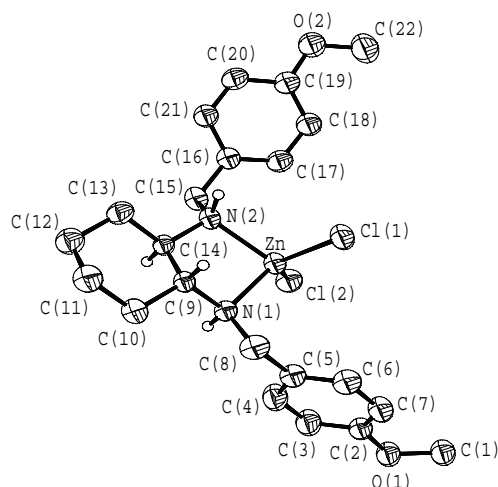
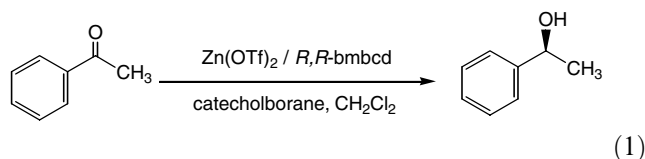


Fig. 1. Crystal structure with the labelling scheme for compound **2**.

in $\text{ZnCl}_2(\text{quinoline})_2$ [21], but larger than that $[81.7(1)^\circ]$ in $\text{ZnCl}_2(\text{Clemizole})$ [20]. The bond angle of $121.42(3)^\circ$ for $\text{Cl}(1)\text{--Zn--Cl}(2)$ is quite similar to that $[120.18(6)^\circ]$ found in $\text{ZnCl}_2(\text{Clemizole})$ [20] and is larger than that $[114.54(7)^\circ]$ in $\text{ZnCl}_2(\text{bdmpame})$ (bdmpame = *N,N*-bis(3,5-dimethylpyrazolylmethyl)-*L*-alaninemethyl ester) [15], that $[114.59(8)^\circ]$ in $\text{ZnCl}_2(\text{MeImOH})_2$ (MeImOH = 2-hydroxymethyl-1-methylimidazole) [23], and that $[115.47(5)^\circ]$ in $\text{ZnCl}_2(\text{quinoline})_2$ [21].

3.3. Catalytic enantioselective reduction of acetophenone

Recently, the enantioselective reduction of ketones prompted by chiral zinc complexes has been well-established, and very high enantioselectivity has been accomplished by using various chiral diimine ligands [11,13]. We have investigated the catalytic enantioselective reduction of acetophenone (Eq. (1)) using in situ $\text{Zn}(\text{OTf})_2/R,R\text{-bmbcd}$ as the catalyst.



The reaction was carried out under the standard set of conditions described in Section 2. The enantiomeric excess of the catalytic reduction was moderate to give 48% ee with 77% yield. The enantioselective reduction of acetophenone using lithiated α -stannylbenzyl sulfide or α -lithio benzyl 2-pyridyl sulfide with *N,N,N',N'*-tetramethyl-cyclohexane-1,2-diamine as the chiral ligand show 1% ee in 8% yield and 21% ee in 70% yield, respectively [24]. The low enantiomeric excess may be explained by the lack of stereoselective control of the functional groups at the reaction center, coupled with the fact that the blocking from the steric hinderance of the methyl or dimethyl groups the near chiral center washes out any chiral discrimination formed in the catalyst.

4. Conclusions

We have attempted to establish the synthesis and characterization of *R,R*-bmbcd and its Zn complex **2**. In particular, the catalytic enantioselective reduction of acetophenone promoted by $\text{Zn}(\text{OTf})_2$ with *R,R*-bmbcd in the presence of catecholborane as reductant has shown that a chiral cyclohexane-1,2-diamine derivative **1** is an exploratory precursor. The zinc catalyst of the new chiral 1,2-diaminocyclohexane derivative ligand, through the steric effect of functional groups and the enhancement of the absolute stereoselective control, could give significant enantiometric excess. We are currently investigating in this area.

5. Supplementary data

Supplementary data are available from the Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK (fax: +44-1223-336033; E-mail: deposit@ccdc.cam.ac.uk or [www: http://www.ccdc.cam.ac.uk](http://www.ccdc.cam.ac.uk)) on request, quoting the deposit number CCDC 231764.

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